

Finding Perpendicular Heights in Parallelograms

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A **parallelogram** is a quadrilateral with two pairs of parallel sides.

Base

The **base** of a parallelogram is any side chosen as the reference side.

Perpendicular Height

The **perpendicular height** (or altitude) is the shortest distance from a base to its opposite side.

The perpendicular height must form a **right angle** with the chosen base.

Area of a Parallelogram

The **area** of a parallelogram is given by

$$A = bh$$

where b is the base and h is the perpendicular height.

Finding the Height

If the area and base are known, the height can be found using

$$h = \frac{A}{b}$$

Finding the Base

If the area and height are known, the base can be found using

$$b = \frac{A}{h}$$

Examples

Example 1

A parallelogram has an area of 120 cm^2 and a base of 15 cm . Find the perpendicular height.

$$h = \frac{A}{b} = \frac{120}{15} = 8 \text{ cm}$$

Example 2

A parallelogram has an area of 156 m^2 and a height of 12 m . Find the base.

$$b = \frac{A}{h} = \frac{156}{12} = 13 \text{ m}$$

Example 3

The area of a parallelogram is 135 cm^2 . If the base is $3x$ cm and the height is 9 cm, find x .

$$135 = (3x)(9)$$

$$135 = 27x$$

$$x = 5$$

Example 4

A parallelogram has a base of 18 m and a height of 13 m. Find its area.

$$A = bh$$

$$A = (18)(13) = 234 \text{ m}^2$$

Example 5

The area of a parallelogram is 240 cm^2 . One base is 16 cm with corresponding height 15 cm. If another base is 24 cm, find its corresponding height.

$$h = \frac{A}{b} = \frac{240}{24} = 10 \text{ cm}$$

Practice Exercises

Practice 1

A parallelogram has an area of 120 cm^2 and a base of 15 cm . Find the perpendicular height.

Practice 2

A parallelogram has an area of 156 m^2 and a height of 12 m . Find the base.

Practice 3

The area of a parallelogram is 135 cm^2 . If the base is $3x$ cm and the height is 9 cm, find x .

Practice 4

A parallelogram has a base of 18 m and a height of 13 m. Find its area.

Practice 5

The area of a parallelogram is 240 cm^2 . One base is 16 cm with corresponding height 15 cm. If another base is 24 cm, find its corresponding height.

Activities

Activity 1

A parallelogram has an area of 90 cm^2 and a base of 10 cm . Find the perpendicular height.

Activity 2

A parallelogram has an area of 108 m^2 and a height of 9 m . Find the base.

Activity 3

The area of a parallelogram is 120 cm^2 . If the base is $x + 3 \text{ cm}$ and the height is 10 cm , find x .

Activity 4

A parallelogram has a base of 16 cm and a height of 9 cm. Find its area.

Activity 5

The area of a parallelogram is 216 cm^2 . One base is 18 cm with corresponding height 12 cm . If another base is 24 cm , find its corresponding height.

Thank you!